

Gradient Estimators and Variance Reduction

Score functions, control variates, importance sampling, Rao–Blackwell, and unbiased sparse credit

Why this tutorial

Hard decisions (which route wins, which memory slot is written) have no ordinary derivative, and exact credit is often too expensive to compute everywhere. Learning then relies on *estimators* of the gradient. The two questions are always: is it *unbiased* (right on average), and what is its *variance* (how noisy)? This tutorial covers the standard toolkit and the specific construction our sparse-credit learners use: a cheap prediction of the credit, corrected by random audits so that it stays exactly unbiased.

1 The score-function estimator

For a distribution $p_\theta(z)$ and a function f ,

$$\nabla_\theta \mathbb{E}_{z \sim p_\theta} [f(z)] = \mathbb{E}_{z \sim p_\theta} [f(z) \nabla_\theta \log p_\theta(z)].$$

Proof: $\nabla \int p f = \int f \nabla p = \int f p \nabla \log p$. One sample z gives the unbiased estimate $f(z) \nabla \log p_\theta(z)$ (REINFORCE). It needs no derivative of f , so it works for discrete choices, but its variance can be large.

Baselines (control variates). Because $\mathbb{E}[\nabla \log p] = 0$, subtracting any constant b keeps it unbiased: $(f(z) - b) \nabla \log p$. For a scalar parameter the variance-minimizing constant is $b^* = \mathbb{E}[f (\nabla \log p)^2] / \mathbb{E}[(\nabla \log p)^2]$.

Key idea

A control variate is anything with known expectation that is correlated with your estimate. Subtracting it costs no bias and removes the shared noise. Better predictions of the quantity give lower variance; a perfect prediction gives zero variance.

2 Importance sampling and self-normalization

To estimate $\mathbb{E}_p[g]$ with samples from a proposal q : $\mathbb{E}_p[g] = \mathbb{E}_q[w g]$ with $w = p/q$, unbiased if $q > 0$ wherever $p g \neq 0$. When p is known only up to a constant (a posterior), one uses *self-normalized* weights $\tilde{w}_s = w_s / \sum_r w_r$. This is biased at finite sample size S (bias $O(1/S)$ under moment conditions) but consistent. With $S = 1$ the normalized weight is always 1: it carries no information about whether the proposal was good.

3 Rao–Blackwellization

If $\hat{g}$ is unbiased and Y is any available information, then $\mathbb{E}[\hat{g} \mid Y]$ is also unbiased and

$$\text{Var}(\mathbb{E}[\hat{g} \mid Y]) \leq \text{Var}(\hat{g})$$

(law of total variance: $\text{Var} \hat{g} = \mathbb{E} \text{Var}(\hat{g} \mid Y) + \text{Var} \mathbb{E}[\hat{g} \mid Y]$). Analytically averaging over anything you can average over never hurts. Example: at the output of a race, use the exact posterior responsibilities ρ_i rather than one sampled winner.

4 Latent causes: Fisher's identity and the missing information

For a model with a latent cause k (which route, which clock), the marginal likelihood gradient is

$$\nabla \log p(y) = \mathbb{E}_{k \sim p(k|y)} [\nabla \log p(y, k)] \quad (\text{Fisher's identity}).$$

So a single *posterior* sample k gives an unbiased estimate. Its covariance is the *missing-information matrix* $\text{Cov}_{k|y}[\nabla \log p(y, k)]$. If every complete-data score has norm at most G and $\rho_k = p(k | y)$,

$$\text{tr Cov} \leq 2G^2 \left(1 - \sum_k \rho_k^2\right) = 2G^2 (1 - e^{-H_2(\rho)}),$$

where H_2 is the collision (Rényi-2) entropy. *Proof sketch:* the variance of a discrete distribution is half its mean squared pairwise distance; each squared distance is at most $(2G)^2$ and pairs with $k \neq l$ have total weight $1 - \sum_k \rho_k^2$. *Reading:* sampled-cause credit is nearly exact when the posterior is sharp.

5 Predicted credit plus sparse audits (unbiased sparse credit)

Let g_b be the exact credit for block b (a parameter block or a state cotangent) and h_b a prediction of it, made before the audit from information already available. Audit block b independently with probability $p_b > 0$ (indicator I_b):

$$\hat{g}_b = h_b + \frac{I_b}{p_b} (g_b - h_b).$$

Proposition 1. $\mathbb{E}[\hat{g}_b] = g_b$ and $\mathbb{E}\|\hat{g}_b - g_b\|^2 = (\frac{1}{p_b} - 1)\|g_b - h_b\|^2$.

Proof. $\mathbb{E}[I_b/p_b] = 1$ gives the mean. The error is $(I_b/p_b - 1)(g_b - h_b)$ and $\mathbb{E}(I_b/p_b - 1)^2 = 1/p_b - 1$. $\square$

The prediction affects only the variance, never the mean. A good predictor makes audits rarely necessary.

Where to audit (Neyman allocation). With audit costs c_b and a variance budget V , minimize $\sum_b c_b p_b$ subject to $\sum_b \|r_b\|^2 / p_b \leq V$, where $r_b = g_b - h_b$ is the *surprise*. By Cauchy-Schwarz,

$$\sum_b c_b p_b \geq \frac{(\sum_b \|r_b\| \sqrt{c_b})^2}{V}, \quad \text{with equality at } p_b \propto \frac{\|r_b\|}{\sqrt{c_b}}$$

(capped at $p_b \leq 1$). Credit cost therefore scales with the *square* of the surprise: heavy while the predictor is poor, cheap once surprises are small, and heavy again if the domain shifts.

Remark. The allocation must use only information available *before* the audit draw (a predicted surprise), and the predictor must be frozen for the current draw. Re-fitting the predictor on the audited value and then recomputing the current baseline breaks the conditioning argument and can bias the estimate.

Exercises

Exercise 1. Show that $\mathbb{E}_{p_\theta}[\nabla \log p_\theta] = 0$.

Solution. $\int p \nabla \log p = \int \nabla p = \nabla \int p = \nabla 1 = 0$.

Exercise 2. In the audit estimator, what happens with $p_b = 1$? With $h_b = g_b$?

Solution. $p_b = 1$: $\hat{g}_b = g_b$ exactly (always audit). $h_b = g_b$: zero variance for any p_b (no surprise).

Exercise 3. Two causes with posterior $(0.9, 0.1)$ and scores bounded by $G = 1$. Bound the variance of single-sample credit.

Solution. $1 - \sum \rho^2 = 1 - 0.82 = 0.18$, so $\text{tr Cov} \leq 0.36$. For $(0.5, 0.5)$ it would be 1.0.

Exercise 4. Derive the Neyman allocation with Lagrange multipliers.

Solution. Stationarity of $\sum c_b p_b + \mu(\sum \|r_b\|^2/p_b - V)$: $c_b = \mu \|r_b\|^2/p_b^2$, so $p_b = \sqrt{\mu} \|r_b\|/\sqrt{c_b}$.

Where this appears in our work

REINFORCE, straight-through and closed-loop baselines in the hard-routing credit diagnostics (note 160 §§3–9); the collision-entropy bound (note 164 Theorem F, Lean-checked); forward-predicted credit with unbiased residual audits (notes 162–163); surprise-squared credit cost (note 165 §10, Lean-checked); Rao–Blackwell conditioning on forward information (note 164 Result 1).